

DANMARKS TEKNISKE UNIVERSITET



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## (02443) Stochastic Simulation

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### PROJECT 2 - HOSPITAL UTILISATION

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An essential element of hospital planning is the decision of how many resources each ward should have. In this project, the aim is to estimate the consequences of different bed distributions and assumptions using simulations, to gain information on what layout is optimal. Specifically, the project will be focusing on a situation in which the hospital has been forced to create two new temporary wards due to a pandemic (Ward A & B), as well as being forced to move staff and beds from an inpatient ward (Ward C). The simulation model will oversee a 365-day period, with the following specifications. The bed distribution cannot be changed during the simulation. The patients' arrivals follow a Poisson process, and their length of stay follows either a log-normal or an exponential distribution. In the first part of the simulation, the hospital consists of 75 beds; however, the effect of more or fewer total beds was later studied.

## Primary Performance Measures

For the first analysis of the primary performance, we've chosen to have 15 beds in A, 10 beds in ward B, and 50 beds in ward C, as well as using a log-normal distribution for the length of stay. The probability that all of the beds are occupied on arrival day for each of the patient types has been estimated using 1000 independent simulation replications. Afterwards, the mean and 95% confidence intervals were found for each ward, which yielded the following results:

$$\text{Ward A : } 0.646, \quad [0.645, 0.648] \quad (1)$$

$$\text{Ward B : } 0.446, \quad [0.444, 0.448] \quad (2)$$

$$\text{Ward C : } 0.292, \quad [0.290, 0.293] \quad (3)$$

This distribution is further visualised in Figure 1. It is observed that the estimated blocking probabilities are quite high, with Ward A being significantly higher than the other two. The graph suggests that for this bed distribution, the temporary wards are at maximum capacity when many patients come in. However, it doesn't say anything about how many are relocated to other hospitals, which is a much more costly outcome

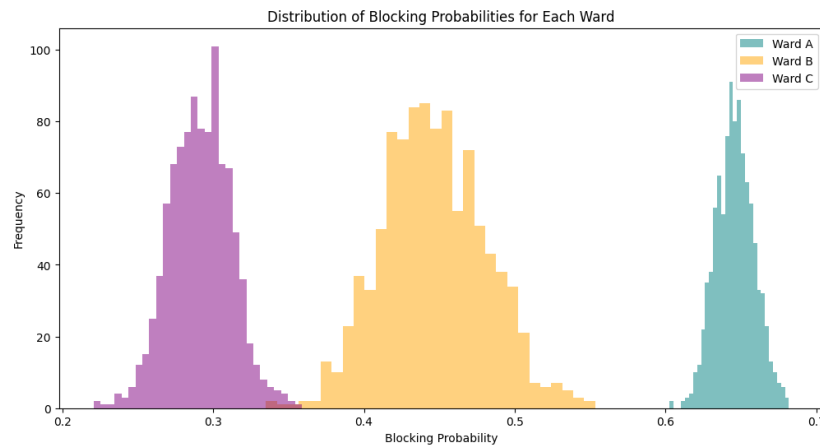


Figure 1: Probability that all beds are occupied on arrival for each of the three patient types with 15 beds in ward A, 10 in ward B and 50 in Ward C

Due to this, the mean number of patients relocated due to a bed shortage was estimated for each patient type. It is observed that the majority of the relocated patients stem from Ward

A, which can be explained by the admission policy, where patients requiring intensive care are admitted to Ward A if all the beds are occupied in Ward B, meaning that Ward A experiences the highest capacity pressure.

$$\text{Ward A : 1435} \quad (\approx 63\%) \quad (4)$$

$$\text{Ward B : 198} \quad (\approx 9\%) \quad (5)$$

$$\text{Ward C : 644} \quad (\approx 28\%) \quad (6)$$

$$\text{Total : 2277} \quad (7)$$

The mean fraction of beds utilised in each ward is  $[0.917, 0.813, 0.965]$ , respectively, indicating that all wards operated at high capacity utilisation throughout the epidemic period. Ward C exhibits the highest average utilisation, suggesting that the creation of temporary epidemic wards placed greater pressure on this ward, despite being the ward with the highest number of beds. Ward A & C experience the highest utilisation due to being used as relocation Wards, compared to Ward B. These results suggest that the hospital operates close to its capacity level throughout the simulated period. To prove this, we have plotted the occupancy level for each day through multiple runs, which can be seen in figure 2

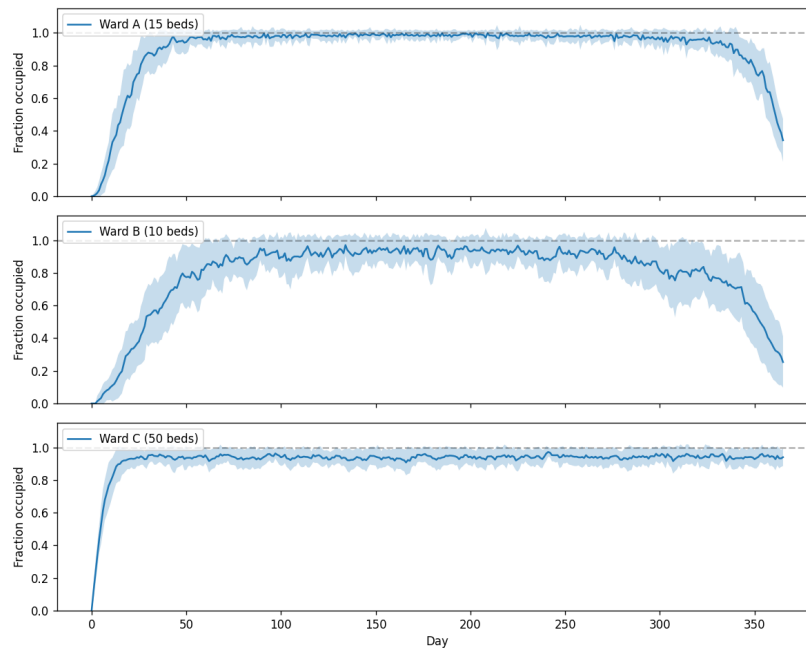


Figure 2: Occupancy of each ward by day, for multiple runs

From this figure we see that it roughly aligns with our expectation based on the mean fraction of beds utilized throughout the entire timeline.

## Sensitivity Analysis

Given the arbitrariness of our initial assumptions (bed distribution and LOS distribution), it is relevant to explore whether a different set of assumptions may impact our findings, and specifically for the bed distribution, whether there is an optimal one.

## Optimal bed distribution

The first part of the project used a randomly selected bed distribution as a baseline. For this part of the project, we attempt to find the optimal bed distribution using a simple grid search to minimise the average number of patient relocations over 100 independent simulations. This optimisation criteria/function was chosen, as was stated before, that relocations to a different hospital can be very costly for both the hospitals and the patients wellbeing.

The optimisation procedure identified the bed distribution (25,5,45) for Ward A, B & C, respectively. The number of rejected patients was  $\approx 2161$  for the one-year simulation period, which remains similar to the parameters chosen at random/initially. Rearranging the beds has a relatively small effect, perhaps due to the fact that regardless of bed arrangement, there is only so much space and after it is taken/filled up, people will have to be relocated. Our optimisation function also doesn't take into account the cost of relocating an intensive care patient vs a regular care patient, which may just cause it to prioritise relocating patients that need more help/will take up a bed for longer, since its still only 1 relocation. This may be why we see in our optimisation that the B ward gets minimized, as if there's space in A they will be moved there anyway, and if there isn't space in A, then someone would have to be relocated anyway. The grid search has been shown visually in figure 3 below

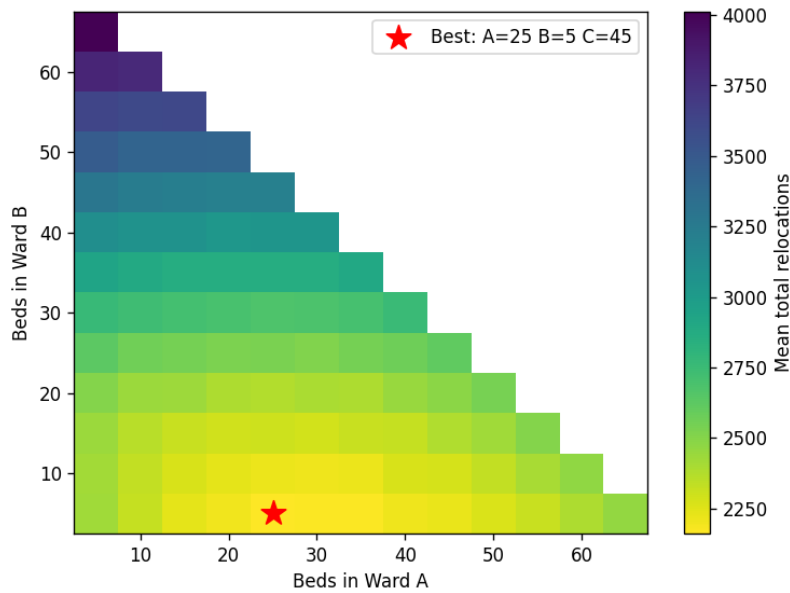


Figure 3: Grid search visualised

The estimated blocking probabilities obtained from 1000 simulation replications were as follows.

$$\text{Ward A: } 0.7317, \quad [0.7310, 0.7324] \quad (8)$$

$$\text{Ward B: } 0.7952, \quad [0.7940, 0.7965] \quad (9)$$

$$\text{Ward C: } 0.1348, \quad [0.1335, 0.1360] \quad (10)$$

The optimised distribution was further illustrated in Figure 4, showing how the probabilities of rejection remain increased for both Ward A & B, but have decreased for Ward C, compared to the randomly selected bed distribution.

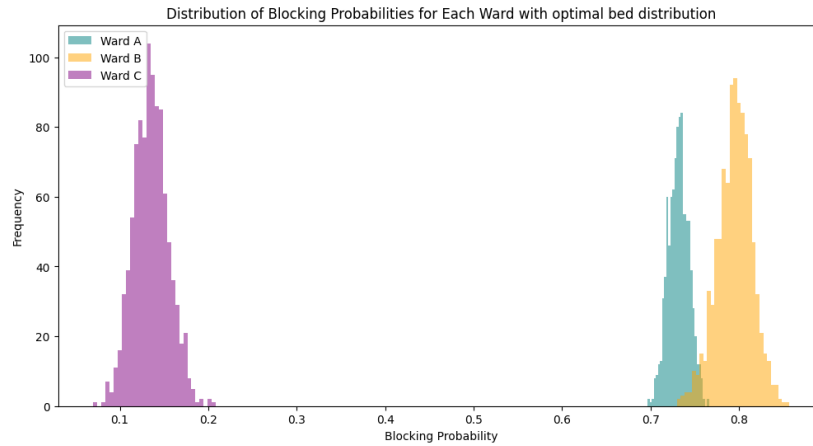


Figure 4: Probability that all beds are occupied on arrival for each of the three patient types using the computed optimised bed distribution

By minimising the number of patient relocations to other hospitals, the optimisation introduces a trade-off where patients face a higher probability of encountering a full warden upon arriving. In particular,  $\approx 80\%$  of intensive care patients are unable to obtain a bed directly in Ward B, which is likely explained by the fact that the optimal solution only allocates 5 beds to Ward B. This creates a higher pressure on this Ward, and is not realistic from a clinical perspective. This is again explained by the objective function minimising total relocations, allowing intensive care patients to relocate to Ward A, and not taking the specialised resources required by intensive care patients into account. Consequently, allocating additional beds to Ward B would reduce the number of beds available in Ward C, which would lead to an overall increase in patient relocation.

Furthermore, the mean fraction of occupied beds was estimated as  $[0.939, 0.900, 0.931]$ , respectively, showing that all three wards operate at more than 90% utilisation on average. The combination of high utilisation and high blocking probabilities suggests that the total capacity of 75 beds is insufficient to accommodate the patient demand generated during the epidemic.

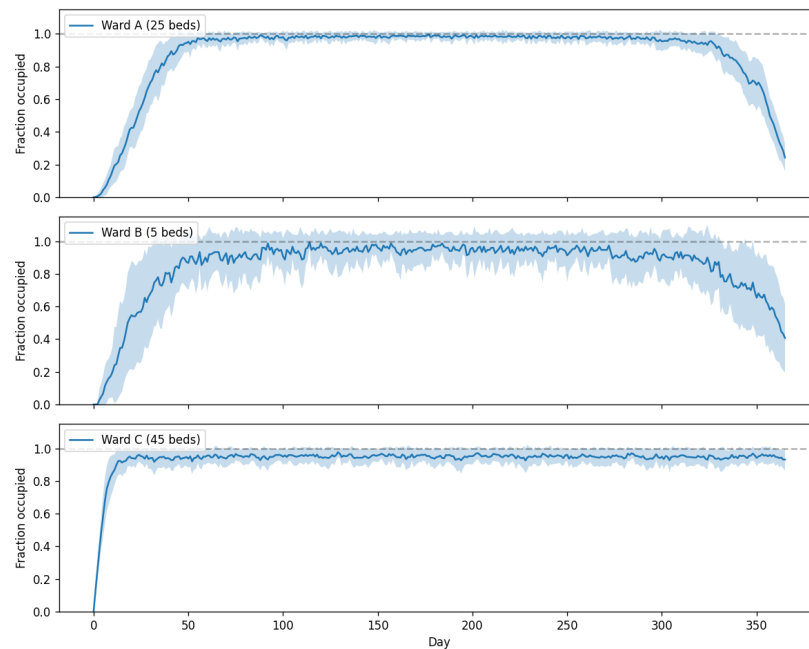


Figure 5: Occupancy over time shown again, but this time for the optimal bed distribution

For the occupancy over time graph as seen before, but now with the optimal bed distribution, we unsurprisingly see a similar graph. The main changes are just slightly higher occupancy all at all times where possible.

## Log-normal vs. Exponential Distribution

When the Length-of-stay distribution is replaced by an exponential distribution with the same mean, the performance metrics change. For the following section the bed distribution was fixed at [15,10,50]. In order to be able to assess the difference, 200 runs of both the regular and the modified simulation were run.

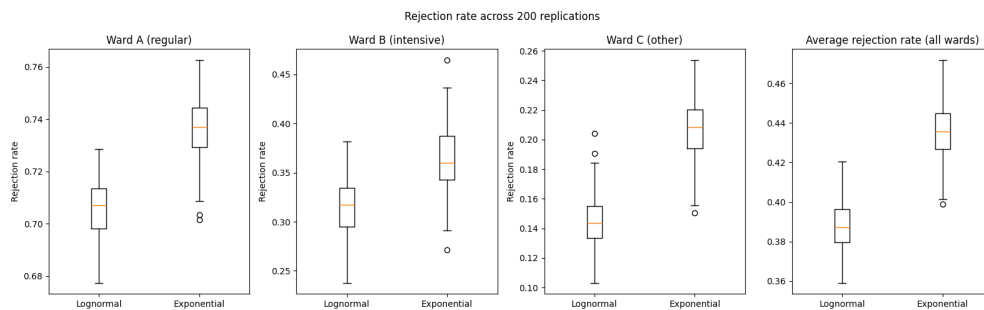


Figure 6: Boxplots showing the difference between the lognormal and exponentially distributed wards, both individually, and as a whole

Figure 6 shows the resulting rejection rates from these runs. These plots indicate that there may in fact be a difference in the rejection rates even if the mean length-of-stay is the same. To

further test this hypothesis statistical tests are needed. First the distribution of the rejection rates for the exponential LOS was plotted in a histogram.

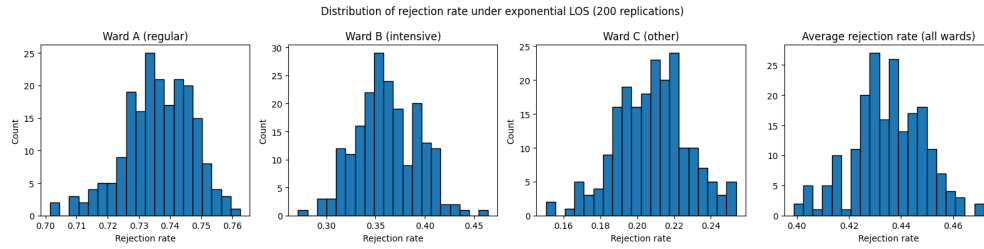


Figure 7: Histograms of per-ward and overall rejection rates using exponential LOS distribution.

Figure 7 shows these distributions. Since they do not appear to be normally distributed, a non-parametric test is needed to compare them to see if they are in fact different. The Mann-Whitney U test allows us to test the null-hypothesis that the actual distributions of the regular and modified hospital rejection rates are in fact the same. The test was carried out and the results were as follows

| Ward    | p-value |
|---------|---------|
| A       | 0.0     |
| B       | 0.0     |
| C       | 0.0     |
| Overall | 0.0     |

Table 1: Mann-Whitney U test for difference in distribution. Note that p values have been rounded, as all are at least 10 orders of magnitude below the critical threshold

The tests clearly show that the change in LOS distribution caused a significant change in the rejection rates.

### The impact of changing the total number of beds with a fixed distribution

Clearly, the easiest solution to the reallocation problem is simply to have more beds. And similarly, if we are interested in saving every bit of money we can, lowering the number of beds may be of interest. Therefore set of new simulations were run where the total number of beds was changed. In order to best study the effect of changing the number of beds, the target ratio was fixed as [3:2:10], although as beds are necessarily whole numbers, this ratio cannot be perfectly captured for all values.

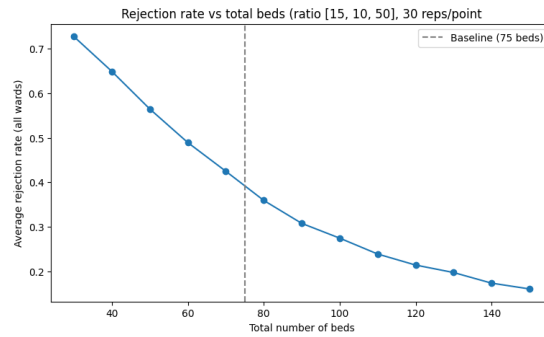


Figure 8: Effect of changing the number of beds, on the total rejection rate

As can be seen in figure 8, the obvious effect is that as the number of beds goes up, the number of rejections goes down. This naturally makes sense, as we can simply fit more patients into the hospital, so if variation results in short inter-arrival or long LOS times we still have space left over.

### The impact of changing the total number of beds with an adaptive distribution

In this section we will shortly explore how our results from the previous section might change given the assumption that the optimal distribution of beds might change depending on the total beds available. To do this, instead of aiming for a target ratio, we run a gridsearch for each datapoint to find the optimal bed distribution for that point. To make this computationally feasible, only 20 independent runs are ran for each point within each grid search, totalling almost 1000 runs. The results can be seen in figure 9

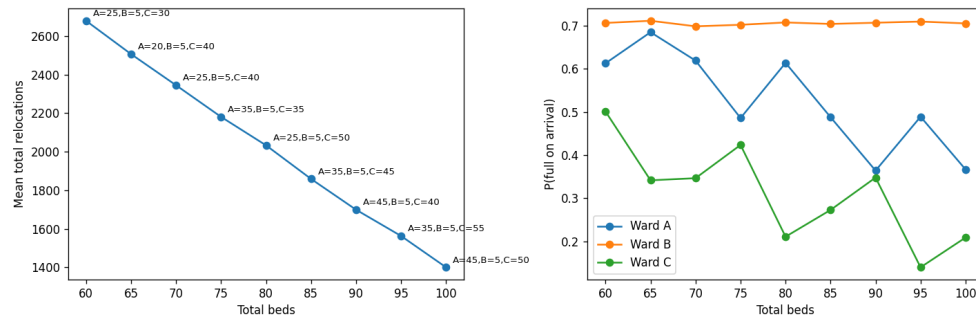


Figure 9: Effect of changing the number of beds, on the total rejection rate and  $p(\text{full})$ , this time adaptively

It is seen on the first graph that using the adaptive distribution, that the total relocations seem to become linear. While this is not impossible to believe given that 1. it only implies a minor improvement on points far away from the original optimization and 2. the optimal relocation rate may be some ratio of incoming patients and the total numbers of beds, it is also not possible to rule out a simulation or assumption error.

Looking at the second graph, we see that the probability of the wards being full when a patient tries to be admitted, slowly falls when given more beds. Furthermore, it almost looks like there



is a tradeoff between optimizing for ward C and optimizing for ward A. While this seems not to be the case when looking at the grid-search distributions, it is an odd artifact.

We see on both plots that again, ward B is always neglected and set to the minimum allotted amount, as the intensive care patients can simply be blocked and moved to A, given space. Perhaps using this information in addition to help from domain experts, a more appropriate optimisation function can be concocted and tested.

## Conclusion

Overall, the simulations showed that the hospital operates under significant capacity pressure, with high bed utilisation and frequent patient relocations across most scenarios. While optimising the bed distribution reduced relocations slightly, the results also highlighted important issues with the optimisation criteria and demonstrated that the total number of available beds has a much greater impact on system performance. These findings illustrate the value of stochastic simulation for evaluating hospital planning decisions

## Use of AI

Throughout this project, AI tools like Claude Code and ChatGPT have been used for assistance with code and debugging.

## Appendix